

AI-Powered DevSecOps CI/CD Platform

Introduction

Modern software development requires fast delivery, secure coding practices, automated testing, intelligent monitoring, and scalable deployment systems. DevOps and DevSecOps practices help organizations automate software workflows while ensuring application security, reliability, and operational efficiency.

The AI-Powered DevSecOps CI/CD Platform was developed to demonstrate modern cloud-native DevOps practices including CI/CD automation, security scanning, monitoring, Kubernetes deployment, and AI-based log analysis. The project integrates multiple DevOps and security tools into a unified automated workflow that simulates a real-world production infrastructure environment.

The platform uses FastAPI for backend API development, Docker for containerization, GitHub Actions for CI/CD automation, Prometheus for monitoring, Kubernetes with Minikube for scalable deployment, and an AI-based log analyzer for intelligent operational monitoring.

Abstract

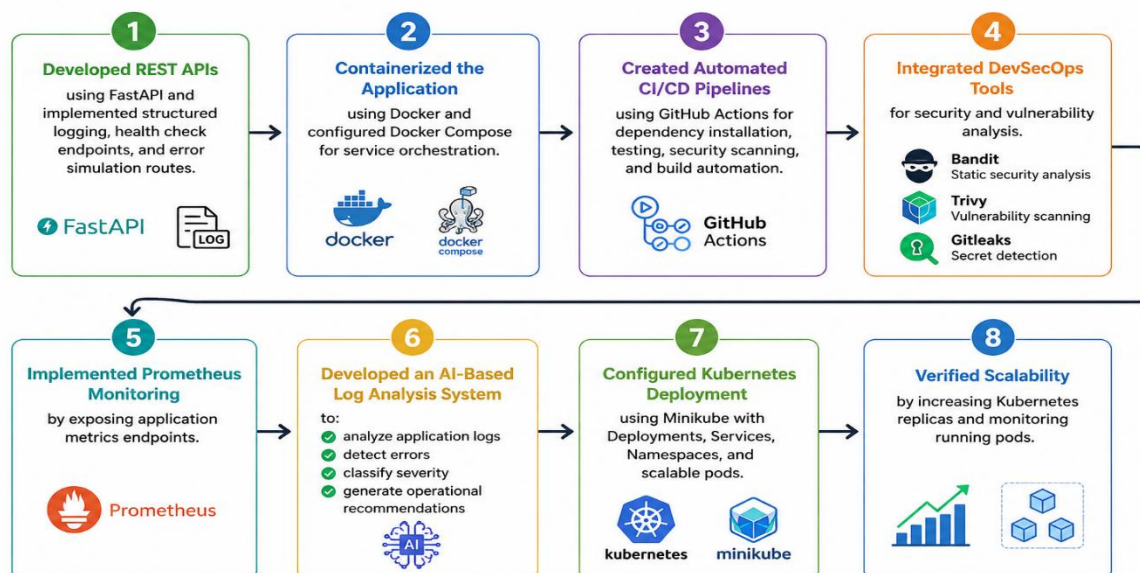
The AI-Powered DevSecOps CI/CD Platform is a cloud-native infrastructure automation project developed to automate application testing, security analysis, monitoring, deployment workflows, and intelligent log analysis. The platform integrates automated CI/CD pipelines, security scanning tools, containerized deployment, monitoring systems, Kubernetes orchestration, and AI-based operational intelligence into a unified workflow that simulates a real-world production environment. The project uses FastAPI for backend API development, Docker for containerization, GitHub Actions for CI/CD automation, Bandit and Trivy for security vulnerability scanning, Gitleaks for secret detection, Prometheus for monitoring, and Kubernetes with Minikube for scalable deployment. An AI-based log analysis module was also implemented to analyze application logs, detect failures, classify severity levels, and generate operational recommendations automatically. Overall, the platform demonstrates modern DevOps and DevSecOps practices commonly used in cloud-native software engineering and infrastructure automation environments.

Tools & Technologies Used

- FastAPI (Backend API Development)
- Python (Programming Language)
- Docker & Docker Compose (Containerization)

- GitHub Actions (CI/CD Automation)
- Bandit (Static Security Analysis)
- Trivy (Vulnerability Scanning)
- Gitleaks (Secret Detection)
- Prometheus (Monitoring & Metrics)
- Kubernetes & Minikube (Container Orchestration)
- Git & GitHub (Version Control)
- Swagger UI / Postman (API Testing)
- AI-based Log Analysis System (Operational Monitoring)

Steps Involved in Building the Project



Conclusion

The AI-Powered DevSecOps CI/CD Platform successfully demonstrates modern DevOps and cloud-native engineering practices including CI/CD automation, containerization, security scanning, monitoring, and Kubernetes orchestration. The project provided practical exposure to real-world deployment workflows and scalable infrastructure management used in modern software systems.

The integration of an AI-based log analysis system enhanced the platform by enabling intelligent error detection, severity classification, and automated operational recommendations. Overall, the project helped build hands-on experience in DevOps, DevSecOps, Cloud, Kubernetes, and AI-assisted operational monitoring relevant to industry-level engineering roles.